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```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.datasets import cifar10
```

```
(train_images, train_labels), (test_images, test_labels) = cifar10.load_data()
train_images = train_images.astype("float32") / 255
test_images = test_images.astype("float32") / 255
class_names = ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']
```

```
model = keras.Sequential([
    layers.Input(shape=(32, 32, 3)),

    layers.Conv2D(64, 3, padding='same', activation='relu'),
    layers.Conv2D(64, 3, padding='same', activation='relu'),
    layers.MaxPooling2D(),
    layers.Dropout(0.3),

    layers.Conv2D(128, 3, padding='same', activation='relu'),
    layers.Conv2D(128, 3, padding='same', activation='relu'),
    layers.MaxPooling2D(),
    layers.Dropout(0.4),

    layers.Flatten(),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(10, activation='softmax')
])
```

```
model.compile(optimizer="adam",
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
model.fit(train_images, train_labels, epochs=15, batch_size=64, validation_split=0.2)
```

```
Epoch 1/15
625/625 ————— 13s 14ms/step - accuracy: 0.8057 - loss: 0.5413 - val_accuracy: 0.7925 - val_loss: 0.6029
Epoch 2/15
625/625 ————— 7s 12ms/step - accuracy: 0.8182 - loss: 0.5097 - val_accuracy: 0.7909 - val_loss: 0.6244
Epoch 3/15
625/625 ————— 7s 12ms/step - accuracy: 0.8184 - loss: 0.5066 - val_accuracy: 0.7971 - val_loss: 0.6021
Epoch 4/15
625/625 ————— 8s 12ms/step - accuracy: 0.8211 - loss: 0.5003 - val_accuracy: 0.7840 - val_loss: 0.6330
Epoch 5/15
625/625 ————— 8s 12ms/step - accuracy: 0.8246 - loss: 0.4885 - val_accuracy: 0.7955 - val_loss: 0.6112
Epoch 6/15
625/625 ————— 7s 12ms/step - accuracy: 0.8294 - loss: 0.4802 - val_accuracy: 0.7874 - val_loss: 0.6389
Epoch 7/15
625/625 ————— 7s 12ms/step - accuracy: 0.8321 - loss: 0.4608 - val_accuracy: 0.7983 - val_loss: 0.6124
Epoch 8/15
625/625 ————— 7s 12ms/step - accuracy: 0.8372 - loss: 0.4594 - val_accuracy: 0.7900 - val_loss: 0.6160
Epoch 9/15
625/625 ————— 7s 12ms/step - accuracy: 0.8350 - loss: 0.4571 - val_accuracy: 0.8009 - val_loss: 0.6102
Epoch 10/15
625/625 ————— 7s 12ms/step - accuracy: 0.8426 - loss: 0.4442 - val_accuracy: 0.8040 - val_loss: 0.5966
Epoch 11/15
625/625 ————— 7s 12ms/step - accuracy: 0.8441 - loss: 0.4374 - val_accuracy: 0.7967 - val_loss: 0.6098
Epoch 12/15
625/625 ————— 7s 12ms/step - accuracy: 0.8500 - loss: 0.4163 - val_accuracy: 0.7982 - val_loss: 0.6252
Epoch 13/15
625/625 ————— 7s 12ms/step - accuracy: 0.8511 - loss: 0.4207 - val_accuracy: 0.8053 - val_loss: 0.6034
Epoch 14/15
625/625 ————— 7s 12ms/step - accuracy: 0.8531 - loss: 0.4132 - val_accuracy: 0.7943 - val_loss: 0.6368
Epoch 15/15
625/625 ————— 7s 12ms/step - accuracy: 0.8504 - loss: 0.4104 - val_accuracy: 0.8032 - val_loss: 0.5980
<keras.src.callbacks.history.History at 0x7c39327ac7a0>
```

```
test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2)
print(f"\nFinal test accuracy: {test_acc:.4f}")
```

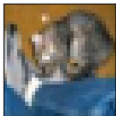
313/313 - 2s - 6ms/step - accuracy: 0.8041 - loss: 0.6302

Final test accuracy: 0.8041

```
import matplotlib.pyplot as plt
import numpy as np
```

```
(train_images_ori, train_labels_ori), (test_images_ori, test_labels_ori) = cifar10.load_data()
test_digits = test_images[0:10]
predictions = model.predict(test_digits)
plt.figure(figsize=(20, 4))
for i in range(10):
    plt.subplot(1, 10, i+1)
    plt.imshow(test_images_ori[i])
    plt.xticks([])
    plt.yticks([])
    predicted_label_index = np.argmax(predictions[i])
    true_label_index = test_labels[i][0]
    predicted_class = class_names[predicted_label_index]
    true_class = class_names[true_label_index]
    color = 'green' if predicted_label_index == true_label_index else 'red'
    plt.xlabel(f"Pred: {predicted_class}\nTrue: {true_class}", color=color)
plt.show()
test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2)
print(f"\nFinal test accuracy: {test_acc:.4f}")
```

1/1 ————— 1s 688ms/step



Pred: dog
True: cat



Pred: ship
True: ship



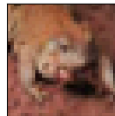
Pred: ship
True: ship



Pred: airplane
True: airplane



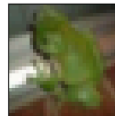
Pred: frog
True: frog



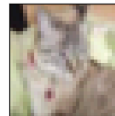
Pred: frog
True: frog



Pred: automobile
True: automobile



Pred: frog
True: frog



Pred: cat
True: cat



Pred: automobile
True: automobile

313/313 - 1s - 3ms/step - accuracy: 0.8041 - loss: 0.6302

Final test accuracy: 0.8041

Start coding or [generate](#) with AI.